

Digital Forensics

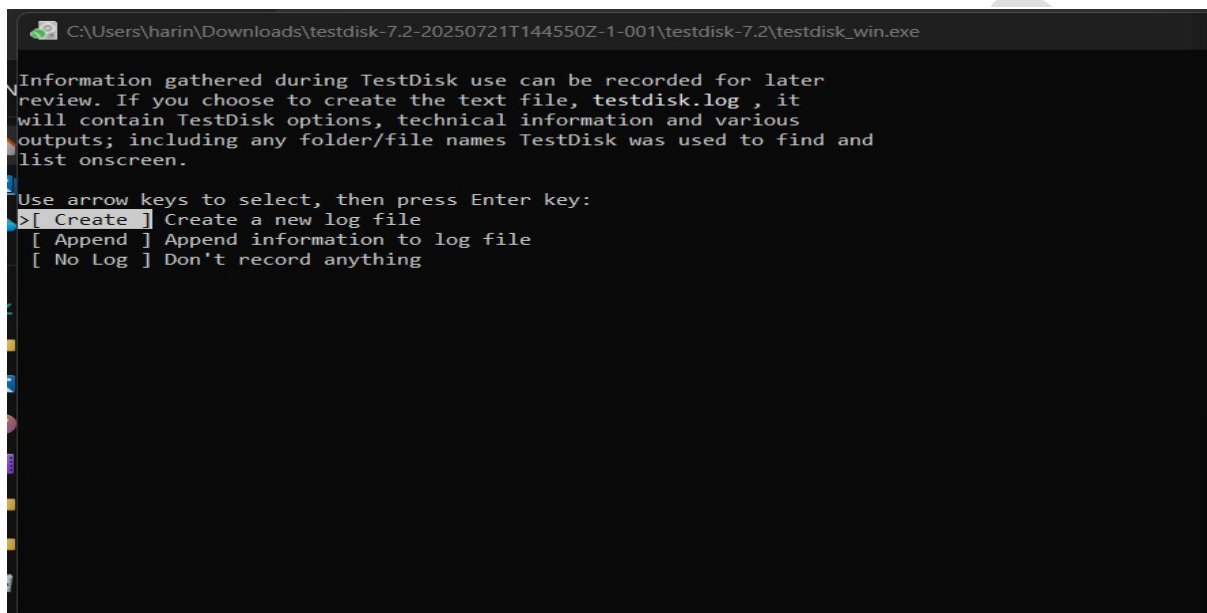
Ex.No:2 Recover deleted or damaged files from a storage device using Test Disk

Date:

Description:

TestDisk step by step to recover a missing partition and repair a corrupted one.

Log creation



```
C:\Users\harin\Downloads\testdisk-7.2-20250721T144550Z-1-001\testdisk-7.2\testdisk_win.exe

Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log , it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
> [ Create ] Create a new log file
  [ Append ] Append information to log file
  [ No Log ] Don't record anything
```

- Choose Create to instruct Testdisk to create a [log file](#) containing technical information and messages, unless you have a reason to append data to the [log](#) or you execute TestDisk from read only media and must create the [log](#) elsewhere.
- Choose None if you do not want messages and details of the process to be written into a [log file](#) (useful if for example Testdisk was started from a read-only location).
- Press Enter to proceed.

Disk selection

All hard drives should be detected and listed with the correct size by TestDisk:

Digital Forensics

```
C:\Users\harin\Downloads\testdisk-7.2-20250721T144550Z-1-001\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - KBG50ZNV512G KIOXIA
>Disk \\.\PhysicalDrive1 - 61 GB / 57 GiB - USB SanDisk 3.2Gen1

>[Proceed] [Quit]
```

Note: Serial number 04010419c03c9a52bb20
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.

- Use up/down arrow keys to select your hard drive with the lost partition/s.
- Press Enter to Proceed.

X If available, use raw device `/dev/rdisk*` instead of `/dev/disk*` for faster data transfer.

Partition table type selection

TestDisk displays the partition table types.

```
C:\Users\harin\Downloads\testdisk-7.2-20250721T144550Z-1-001\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 61 GB / 57 GiB - USB SanDisk 3.2Gen1

Please select the partition table type, press Enter when done.
>[Intel] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax] Humax partition table
[Mac] Apple partition map (legacy)
[None] Non partitioned media
[Sun] Sun Solaris partition
[XBox] Xbox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

- Select the partition table type - usually the default value is the correct one as TestDisk auto-detects the partition table type.
- Press Enter to Proceed.

Current partition table status

TestDisk displays the menus (also see [TestDisk Menu Items](#)).

Digital Forensics

```
C:\Users\harin\Downloads\testdisk-7.2-20250721T144550Z-1-001\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 61 GB / 57 GiB - USB SanDisk 3.2Gen1
CHS 7479 255 63 - sector size=512

[ Analyze ] Analyze current partition structure and search for lost partitions
[ Advanced ] Filesystem Utils
[ Geometry ] Change disk geometry
[ Options ] Modify options
[ MBR Code ] Write TestDisk MBR code to first sector
[ Delete ] Delete all data in the partition table
[ Quit ] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyze'
process may give some warnings if it thinks the logical geometry is mismatched.
```

- Use the default menu "Analyze" to check your current partition structure and search for lost partitions.
- Confirm at Analyze with Enter to proceed.

Now, your current partition structure is listed. Examine your current partition structure for missing partitions and errors.

```
C:\Users\harin\Downloads\testdisk-7.2-20250721T144550Z-1-001\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.\PhysicalDrive1 - 61 GB / 57 GiB - CHS 7479 255 63
Current partition structure:
Partition      Start      End      Size in sectors
-----
1 P FAT32 LBA    0    0 33  7479 225 42  120164320 [NO NAME]
No partition is bootable

*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
>[Quick Search] [ Backup ] Try to locate partition
```

The first partition is listed twice which points to a corrupted partition or an invalid partition table entry.

Invalid NTFS boot points to a faulty NTFS boot sector, so it's a corrupted filesystem.

Only one logical partition (label Partition 2) is available in the extended partition. One logical partition is missing.

- Confirm at **Quick Search** to proceed.

Quick Search for partitions

Digital Forensics

TestDisk displays the first results in real time.



(click on thumb to display the image).

During the *Quick Search*, TestDisk has found two partitions including the missing logical partition labeled *Partition 3*.

```
C:\Users\harin\Downloads\testdisk-7.2-20250721T144550Z-1-001\testdisk-7.2\testdisk_win.exe
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.PhysicalDrive1 - 61 GB / 57 GiB - CHS 7479 255 63
Current partition structure:
  Partition      Start      End      Size in sectors
  1 P FAT32 LBA   0  0 33  7479 225 42  120164320 [NO NAME]
No partition is bootable

TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.PhysicalDrive1 - 61 GB / 57 GiB - CHS 7479 255 63
  Partition      Start      End      Size in sectors
  1 * FAT32 LBA   0  0 33  7479 225 42  120164320 [NO NAME]
```

- Highlight this partition and press **p** to list your files (to go back to the previous display, press q to Quit, Files listed in red are deleted entries).

All directories and data are correctly listed.

- Press Enter to proceed.

Save the partition table or search for more partitions?

```
C:\Users\harin\Downloads\testdisk-7.2-20250721T144550Z-1-001\testdisk-7.2\testdisk_win.exe
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk \\.PhysicalDrive1 - 61 GB / 57 GiB - CHS 7479 255 63
  Partition      Start      End      Size in sectors
  1 * FAT32 LBA   0  0 33  7479 225 42  120164320 [NO NAME]

[ Quit ] [ Return ] >[Deeper Search] [ Write ]
Try to find more partitions.
```

- *When all partitions are available* and data correctly listed, you should go to the menu **Write** to save the partition structure. The menu **Extd Part** gives you the

Digital Forensics

opportunity to decide if the extended partition will use all available disk space or only the required (minimal) space.

```
C:\Users\harin\Downloads\testdisk-7.2-20250721T144550Z-1-001\testdisk-7.2\testdisk_win.exe
https://www.cgsecurity.org

Disk \\.PhysicalDrive1 - 61 GB / 57 GiB - CHS 7479 255 63
Current partition structure:
  Partition      Start      End      Size in sectors
1 P FAT32 LBA    0  0 33  7479 225 42  120164320 [NO NAME]
No partition is bootable

TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

You will have to reboot for the change to take effect.

[Ok]
```

- TestDisk displays *You have to restart your Computer to access your data* so press Enter a last time and reboot your computer.